Class Topper Finder

Week 9 - Category B Coding Assignment

Problem and approach

Return (rowIndex, total) for the student with the highest total marks. Updating only on a strictly greater sum preserves the first row when totals tie.

Solution (Python)

```
def findTopper(marks):
    best_index = 0
    best_total = -1

    for row_index, row in enumerate(marks):
        total = 0
        for mark in row:
            total += mark
        if total > best_total:
            best_total = total
            best_index = row_index

    return (best_index, best_total)

marks = [[78, 85, 90],
         [88, 92, 79],
         [65, 70, 95]]
print(findTopper(marks))

# Tie example: strictly greater preserves the earlier row.
print(findTopper([[50, 50], [60, 40], [20, 30]]))
```

Output

```
(1, 259)
(0, 100)
```

Complexity

Time: O(m * n), because each of the m students and n marks per student are visited once. Extra space: O(1), excluding the input grid.

Correctness note

The total of every row is calculated exactly once. The stored answer changes only when a larger total is found, so equal totals retain the smallest row index.